

Dr. Babasaheb Ambedkar Technological University Lonere

Semester Examination

S. Y. B. Tech. (Mechanical)

Subject: Machine Drawing and Computer Aided Drafting

Max. Marks: 60

Max. Time: Four hrs.

INSTRUCTIONS: 1. Attempt any **FIVE** questions.

2. Necessary data are given in the respective questions. If such a data is not given, it means that the knowledge of the data is a part of the examination.

3. Make suitable assumptions if necessary and state them clearly giving reasons.

Que.1: A pictorial view of a clamp is shown in Figure A. Draw full size the following views by using first angle of method of projection.

a) Front view looking in the direction of X.

b) Top view

c) Half sectional side view (Section along A-B) (12)

Que.2:

A) Draw symbols for the following pipe fittings and valves. (4)

a) cross b) cap c) gate valve d) plug

B) Draw neat and dimensioned sketch of the following (ANY TWO) (8)

i) Eye foundation bolt

ii) Square and ACME thread profile

iii) Castle nut

iv) Knuckle or Pin joint

Que.3: Two unequal size pipes, Main pipe vertical and 88 mm diameter and branch pipe inclined at 60° to vertical and of 50 mm diameter, are connected with their axes offset and parallel to V. P. Axes of branch pipe is nearer to V.P. by 15 mm as compared to the axis of the vertical main pipe. draw the projections along with curve of penetration or intersection. (12)

Que.4:

Figure B shows two views of Universal coupling for 50 mm diameter shafts. Material list of the coupling is as given below. Draw two orthographic views of each coupling details. (12)

S. No.	Part Name	Material	Quantity
1	Fork	CI	2
2	Center block	CI	1
3	Pin	MS	2
4	Taper pin	MS	2
5	Collor	MS	2

Que.5:

A) Determine allowance and tolerances for the following dimensions of mating parts according to hole basis system. State the type of fit. (04)

Shaft	Hole
Ø29.90 mm	Ø 30.000 mm
Ø29.85 mm	Ø30.025 mm

B) Define the following terms related to surface roughness. (04)

- | | |
|-----------------------------|------------------|
| a) Waviness | b) Roughness |
| c) Surface roughness number | d) Ideal surface |

C) What do you understand by lay? Draw various type of lays with their symbols. (04)

Que.6:

A) Explain in brief the followings

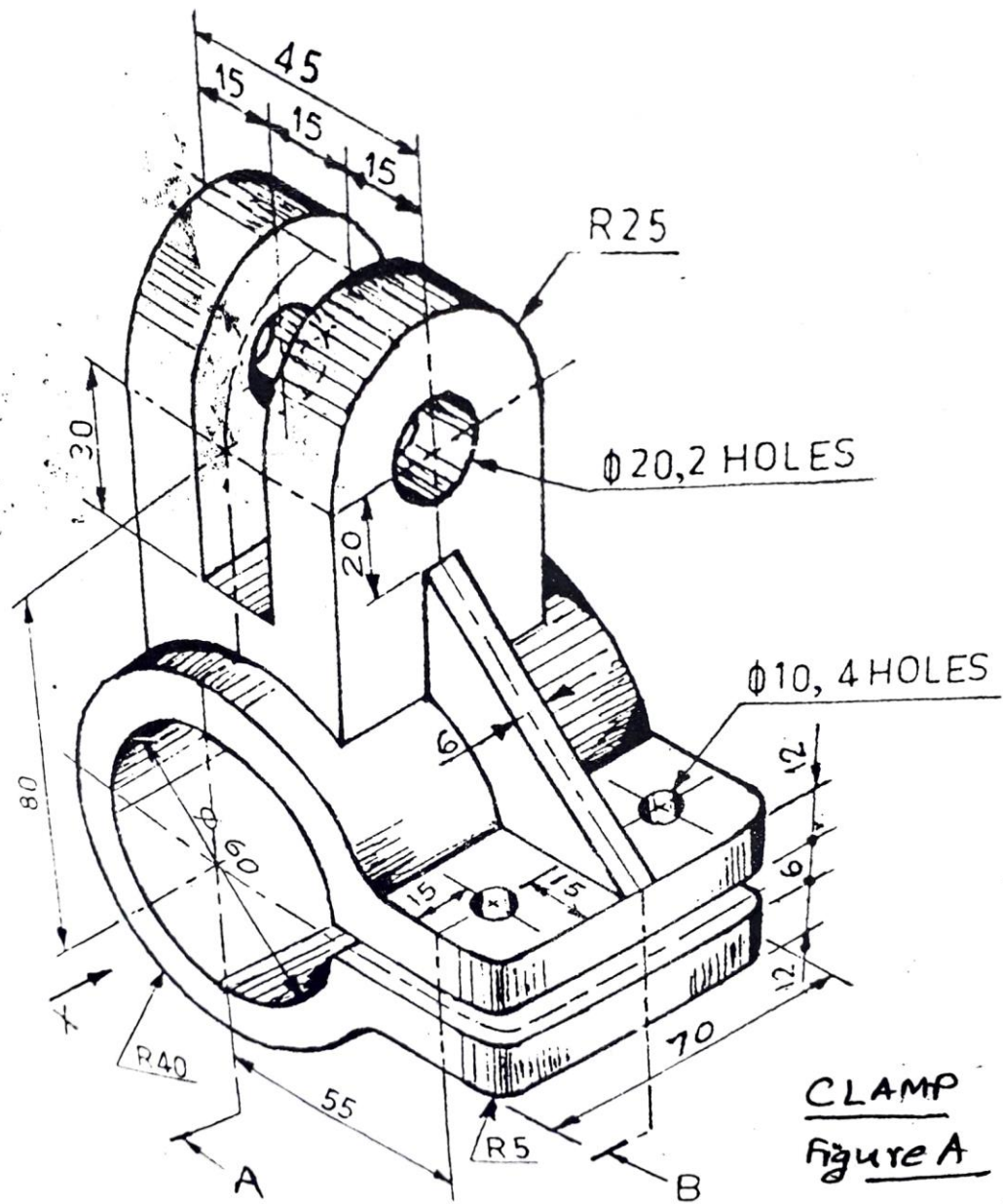
- | | |
|------------------------|------|
| i) Wire frame modeling | |
| iii) Solid modeling | (04) |

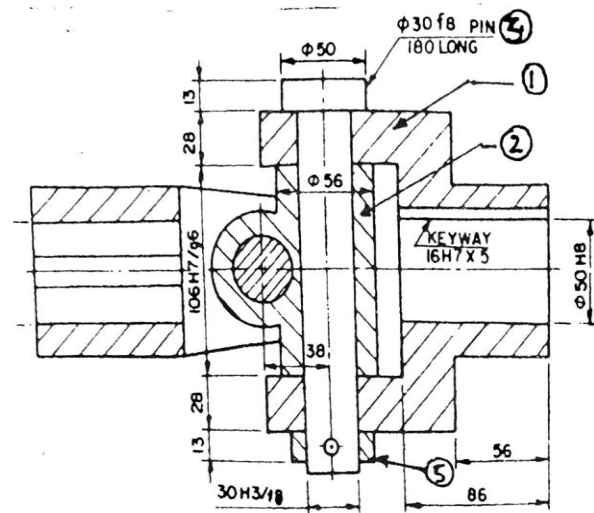
B) Explain the following 2D drawing entities with help of sketch.

- | | | | | |
|--------|-----------|--------------|--------------|------|
| i) Arc | ii) Donut | iii) Polygon | ii) Polyline | (04) |
|--------|-----------|--------------|--------------|------|

C) Explain the boolean operations that can be performed on solid primitives with suitable example. (04)

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Sr No	Part's Name	Material	No off
1	Fork	C I	2
2	Center block	C I	1
3	Pin	M S	2
4	Taper pin	M S	2
5	Collar	M S	2

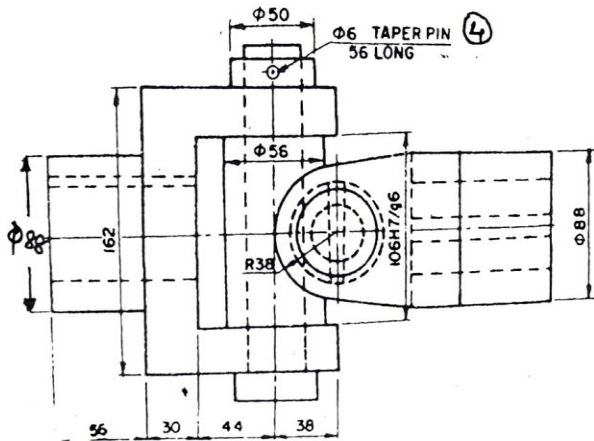


Figure B